Fiscal Year:	FY 2011	**Task Last Updated:**	FY 03/04/2011
PI Name:	Dickman, J. David Ph.D.		
Project Title:	Altered Gravity Effects Upon Vestibular Adaptation, Development and Regulation of Bone Homeostasis		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	77030-3411	**Congressional District:**	7
Comments:	NOTE PI moved to Baylor in Fall 2011; previously at Washington University, St. Louis, MO (Ed., 12/20/2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	02/01/2010	**End Date:**	01/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ornitz, David (Washington University)		
Grant/Contract No.:	NNX10AE38G		
Performance Goal No.:			
Performance Goal Text:			
	Exposure to microgravity or hypergravity environments provides new adaptive challenges for many of the body's major systems. The primary objective of the current collaborative project is to examine the effects of exposure to altered gravity conditions upon vestibular receptor morphology and regulatory mechanisms of bone homeostasis. A number of established and recent investigations have shown that exposure to altered gravity conditions can significantly affects all of these processes. Examining the effects of altered gravity conditions upon vestibular receptor formation is possible using normal animals placed in either hypergravity or appropriately controlled rotational 1g environments. Yet, investigations regarding vestibular regulatory functions are multivariate and difficult to interpret. However, the recent development of transgenic mice with phenotypes that either lack the ability to sense gravity conditions or are completely normal has provided new investigative tools. Here, heterozygous Otopetrin mice (Otop1 ßgal+) that harbor a ßgal		

Task Description:	reporter gene and exhibit a normal otoconia phenotype, as well as homozygous Otopetrin deficient mice (Otop1 ßgal/tlt) exhibit a complete absence of vestibular receptor otoconia will be used. Otop1 ßgal/+ mice are hypothesized to sense changes in gravity load that result in altered vestibular function and an increase in vestibular regulation of bone remodeling when exposed to long-duration hypergravity (2g). In contrast, Otop1 ßgal/tlt mice, where no vestibular response to changes in gravity load are detectable, are hypothesized to exhibit no relative change due to altered gravity exposure as compared to heterozygous controls. In fact, use of the heterozygous and homozygous Otopetrin mice offers an ideal control for examining vestibular regulatory mechanisms in response to changes in gravity conditions, since other effecting variables, such as stress and skeletal load, will be identical for the two groups of mice. In the proposed studies, we will examine the effects of altered gravity exposure upon vestibular receptors for both adaptation and development, using hypergravity and rotational 1g conditions. We will investigate effects upon vestibular receptor formation by examining receptor cells, afferent nerve innervation, Otopetrin expression, and otoconia synthesis. We will investigate the effects upon vestibular regulation of bone homeostasis by examining bone density measures of the forelimb and hind limbs using micro computerized tomography. These studies promise to provide crucial information into the role of gravity as a determinant for vestibular system development, adaptation, and regulation of other major systems. Before long-term space missions are manifested for habitation on other planets, moons, or environments where altered gravity conditions are certain, it is crucial to understand how the vestibular system will adapt in structure, function, and regulatory behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of the vestibular system that coordinates all of our movement, stabilizes our gaze, and provides directional information for spatial orientation is highly dependent upon the presence of gravity. In fact, gravity has been shown to be powerful determinant for development in many systems, including the nervous, muscular, and skeletal systems. Our research will directly assess how these systems utilize information derived from vestibular receptors as epigenetic determinants for normal system development, as well as adaptation to altered gravity environments. When vestibular receptors are compromised, what are the consequences to normal development? Our work will provide insights regarding vestibular receptors and bone formation in normal animals and those exposed to altered gravity conditions, such as may be encountered during space habitation, or growth on extraterrestrial bodies.
Task Progress:	In the first budget year, we have concentrated upon examination of the development of otoconia and the expression patterns of two genes important for otoconial formation, Otoconin 90 (OC90) and Otopetrin 1 (Otop1). Since we had originally worked out the protocols for examination of these products in quail, we finished that project before proceeding onto working with mice. We have a paper under review entitled "Kim, E., Huss, D., Ignatova, E., Thalmann, R., Thalmann, I., Ornitz , D.M., and Dickman, J.D. Development of otoconia and expression of Otopetrin 1 and Otoconin 90 in quail raised in normal and hypergravity environments" regarding the findings.
Bibliography Type:	Description: (Last Updated: 08/06/2012)
Articles in Peer-reviewed Journals	Kim E, Huss D, Ignatova E, Thalmann R, Thalmann I, Ornitz DM, Dickman JD. "Development of otoconia and expression of Otopetrin 1 and Otoconin 90 in quail raised in normal and hypergravity environments." Hearing Research. (under review), January 2011. , Jan-2011